Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/06/2025
PI Name:	Taylor, W Robert M.D., Ph.D.		
Project Title:	Potential Role of the Endothelium in Internal Jugular Venous Thrombosis due to Abnormal Venous Flow Patterns During Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	11/13/2020	**End Date:**	11/01/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/01/2024 per NSSC information (Ed., 1/13/24) NOTE: End date changed to 11/12/2023 per NSSC information (Ed., 11/16/22) NOTE: End date changed to 11/12/2022 per NSSC information (Ed., 11/18/21)		
Key Personnel Changes/Previous PI:	Patrick Jones, a MD/PhD Student in the joint Emory/Georgia Tech BME program has joined the project.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC21K0251		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	The recent report of complete and partial thrombosis of the internal jugular vein in crew members on the International Space Station (ISS) raises concerns for the potential of life threatening pulmonary emboli. There appear to be very abnormal blood flow patterns in the internal jugular vein in zero gravity that may predispose to local thrombus formation. Given that the endothelium is uniquely sensitive to changes in the local flow/shear stress environment, the focus of this proposal is on the potential role of the endothelium in mediating localized thrombus formation. The purpose of this proposal is to identify changes in gene expression in venous endothelium exposed to the same flow patterns as those observed in the ISS crew members using isolated vein organ culture system. These studies will help us to develop a better understanding of the basic mechanisms responsible for thrombosis formation with the ultimate goal of potentially identifying biomarkers that would enable screening and risk stratification of crew members.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will help us to better understand the fundamental physiologic impact of altered flow patterns on the venous endothelium. In addition to allowing us to better understand the differences between venous and arterial endothelial responses, the work may have direct applications to helping us to understand the pathophysiology of venous disease. Finally, venous grafts are used in peripheral and coronary artery bypass surgery and we know that they have a limited, useful lifespan. Given that these segments are placed in a flow environment that is different from the natural flow environment, the work in this proposal may provide insight into how venous endothelium responds to non-physiologic flow stimuli.
Task Progress:	As detailed in previous progress reports, the major technical challenge that emerged was the inability to harvest enough viable endothelial cells from the isolated vein segments in order to isolate sufficient amounts of mRNA for sequencing. We explored multiple different cell isolation cocktails using both human and mouse vein segments without success. Therefore, we switched to cell culture based system using cultured human venous endothelial cells. Published mean blood velocities, and geometric data from Doppler ultrasound acquired in astronauts on the International Space Station (ISS), were used to calculate wall shear stress (WSS) values for normal and thrombus-associated flows, yielding a 4 dyne/cm2 steady flow condition and ± 2 dyne/cm2 oscillatory flow condition respectively. Human primary saphenous and inferior vena cave vein endothelial cells were exposed to these WSS conditions for 24 hours in a commercial parallel plate flow apparatus (ibidi Pump System). For comparison, human primary aortic endothelial cells were also studied given the many publications using this cell type. The most surprising initial result was that we observed differences in the alignment of the two cell types to flow. It is has been well established that aortic endothelial cells from many species align with unidirectional laminar flow, a finding that is confirmed in vivo. However, we found that while both cell types were not aligned in the oscillatory flow pattern (as expected), under unidirectional laminar flow, the venous endothelial cells actually aligned perpendicular to the direction of flow. We performed some initial RT-PCR analyses of the venous endothelial cells to demonstrate feasibility as well as to evaluate the expression of specific pre-inflammatory and pro-thrombotic candidate genes. The candidate genes were selected based on previously published reports using aortic endothelial cells. We found at these low levels of shear stress, that the expression o ICAM-1, VCAM-1, and MCP-1 was not different between steady or oscillatory shear stress in either cell type. We also examined the expression of NOX4, a known component of the NADPH oxidase system in endothelial cells, and found that the expression of it too was not different between the two flow conditions. Interestingly the expression of KLF2, a transcription factor known to be a major regulator of shear stress responses in the endothelium, was upregulated under conditions of steady flow which is consistent with previously published reports. Also of interest was the observed increase in baseline expression of NOX4 in the venous endothelial cells compared to the aortic endothelial cells, suggesting fundamental differences in redox mechanisms. Despite the end of funding, we are continuing this work. Currently, we are performing additional cell exposure studies in order to examine a panel of cardiovascular and thrombotic genes.
Bibliography Type:	Description: (Last Updated:)